University of Pisa
Department of Physics
Master's Degree in Physics of Complex Systems

Anomalous Transport in Soaring Flights

Collective Strategies and Intermittent Search Models

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Abstract

Soaring birds and human pilots stay aloft by extracting energy from atmospheric updrafts, producing long, intermittent trajectories that alternate climbs within thermals with quasi-ballistic glides. Such motion is a candidate for *anomalous transport*, that is, a departure from ordinary Brownian diffusion. This thesis characterizes the transport statistics of real soaring flights from a large empirical dataset and compares them with stochastic transport models.

This document reports the current state of the project. A first, reproducible acquisition has collected 203,343 flights spanning 27 seasons (1999–2000 to 2025–2026) from the FFVL Coupe Fédérale de Distance, of which 186,225 carry a GPS tracklog and 186,052 have been downloaded and verified. Data collection is ongoing and may be extended to further public soaring databases (for example XContest), using the same scraping approach. The remainder of the document describes the acquisition method, the dataset and its statistics, and the planned analysis and modelling.

Contents

Chapter 1

Introduction

1.1 Motivation

Gliders, paragliders, and soaring birds carry no continuous source of propulsion; to remain airborne they extract energy from the moving atmosphere, primarily from *thermals*, the buoyant updrafts that form as sunlight heats the ground. A cross-country flight thus proceeds as a repeating cycle of three phases. In the *transition* phase the flyer glides along a nearly straight, energy-efficient path toward the next source of lift. If none is reached at a safe altitude, a *search* phase follows, during which the flyer probes the surrounding air mass to locate lift. Once lift is found, a *climb* phase begins, in which the flyer circles within the rising air to regain altitude. The cycle alternates long, directed displacements with episodes of slow, localized motion.

Intermittent and directionally persistent motion of this kind is the setting in which *anomalous transport* arises. Let $r(t)$ denote the horizontal position of a flight. Transport is anomalous when the mean-squared displacement grows as a non-trivial power of time, $\langle r^2(t)\rangle \sim t^\alpha$ with $\alpha \neq 1$, as opposed to the linear growth ($\alpha = 1$) of ordinary Brownian diffusion. Continuous-time random walks and Lévy walks are the standard frameworks for such regimes [1, 2], which have been reported in animal movement, in disordered media, and in turbulent flows. Whether the trajectories of real soaring flights belong to an anomalous regime, and of which type, is the question addressed in this thesis.

Soaring has previously been studied mainly as a control problem—how to centre and exploit a thermal efficiently, including by reinforcement learning [3]. The present work is statistical instead: it asks which transport properties emerge at the level of the ensemble of observed trajectories. Its direct precedent is the study of Vilpellet et al. [4], who analysed a large set of human soaring flights, found a horizontal transport that is sub-ballistic and approximately independent of aircraft type (Hurst exponent $H \approx 0.88$), and introduced a phase-resolved analysis based on the transition–search–climb cycle. This thesis builds on that study: it enlarges the dataset and re-examines both the phase segmentation and the transport analysis.

1.2 Objectives

The thesis pursues three objectives:

1. to assemble a large, reproducible, and documented dataset of real soaring flights;

2. to reconstruct clean trajectories from the raw GPS logs and define suitable transport observables;

3. to test for anomalous transport and to compare the empirical statistics with stochastic transport models.

A first dataset—the FFVL CFD archive—has been assembled and verified and is described here; data collection is ongoing and may incorporate further sources (Chapter 2). The second and third objectives define the work ahead (Chapter 4).

1.3 Scope of this document

This document reports the current state of the project and is revised as the work proceeds. Its quantitative content—the summary statistics and tables—is generated directly from the dataset, so it remains consistent with the data in hand. Chapter 2 describes the acquisition of the data and the reproducibility of the pipeline; Chapter 3 describes the dataset and its statistics; and Chapter 4 sets out the planned analysis and modelling.

Chapter 2

Data acquisition

2.1 Source: the FFVL Coupe Fédérale de Distance

The data come from the *Coupe Fédérale de Distance* (CFD), the cross-country competition run by the French free-flight federation (FFVL). Pilots upload their flights; each entry carries metadata (pilot, date, take-off and landing sites, scored distance, duration, glider) and, in most cases, the raw GPS tracklog in IGC format. The competition publishes one public list per season, addressable as a machine-readable XML export. A “season” is identified by its starting year, e.g. the year 1999 denotes the 1999–2000 season.

2.2 What is collected

For every season the pipeline retrieves two things:

- the **season XML export**, archived verbatim as provenance (`raw_xml/<year>.xml`); it is the authoritative list of flights and links;
- the **IGC tracklog files** referenced by the XML, one file per flight that has a track.

From these, a derived **catalog** is built (`catalog.csv`, one row per flight, metadata plus the local path of its track) together with a per-season index (`seasons_index.csv`). The raw data (tens of gigabytes) live on an external disk and are never committed to the repository; only the small per-season index is versioned, and it is the source of the statistics in Chapter 3.

2.3 Pipeline and tooling

Acquisition is performed by a dedicated, installable Python package (`soaring.acquisition.ffvl`) exposed through a command-line tool, `soaring-ffvl`, with five subcommands: `fetch-xml` (archive the season XMLs), `download` (fetch the IGC files), `build-catalog` (regenerate the catalog and the index), `status` (per-season coverage), and `verify` (on-disk integrity check). The design favours robustness for a long-running, large job:

Resumable downloads. Files already present are skipped, so the job can be interrupted and resumed freely.

Atomic writes. Each file is written to a temporary `.part` and then renamed, so a file under its final name is always complete – important on the exFAT external disk, which has no journaling.

Content validation. A response is accepted only if it looks like a real IGC file (an `A` header record followed by at least one `B` fix record); HTML error pages and truncated responses are rejected and retried.

Courteous networking. A small bounded thread pool, per-request timeouts, exponential backoff, and a minimum delay between requests avoid hammering the server; a browser TLS fingerprint is impersonated to pass the site's Cloudflare challenge.

2.4 Reproducibility

The destination directory (`data_root`) is not hard-coded: it is set per machine through an environment variable (`SOARING_FFVL_DATA_ROOT`) or a configuration file shipping a placeholder, so no machine-specific path is committed. Every command validates this setting at start-up and aborts with an explicit message if the target is missing, rather than failing obscurely. Archiving the season XMLs verbatim keeps the full provenance: the catalog can always be rebuilt from them with `build-catalog`.

2.5 Integrity

Because the data sit on an external drive, the `verify` subcommand re-scans every downloaded file on disk, reusing the same IGC validation applied at download time, and flags truncated or invalid files, leftover `.part` fragments, and read errors. The FFVL collection has been verified in full: all 186,052 downloaded tracklogs pass validation (Chapter 3).

2.6 Further data sources

The FFVL CFD is the first and, at present, the primary source, but the dataset is not assumed to be final. Additional public soaring repositories are being pursued—notably XContest, and possibly others—to enlarge and diversify the sample. These repositories expose comparable flight metadata and IGC tracks, so they can be acquired with the same scraping approach and merged into the same catalog. The analysis is designed to treat the resulting collection uniformly, stratifying by source where this matters.

Chapter 3

The dataset

3.1 The IGC tracklog format

A flight track is stored as an IGC file, the standard plain-text format of the Fédération Aéronautique Internationale for flight recorders. Each line is a typed record whose *type is fixed by its first letter*, a code defined by the standard rather than chosen by the logger. The relevant codes are:

`A` the logger/manufacturer identifier (first record);

`H` header records (date of flight, pilot, glider, datum, . . .);

`I` the declaration of optional extra fields appended to each fix;

`B` the GPS fixes – the core of the file; "`B`" is simply the code the standard reserves for fix records.

A `B` record packs its fields by fixed character position. For example,

```
B 094329 4933044N 00005193E A 00095 00149
```

reads as: UTC time `09:43:29`; latitude 49°33.044′N; longitude 000°05.193′E; a fix-validity flag (`A` for a valid 3-D fix, `V` for a 2-D one); and then *two* altitudes, the barometric (pressure) altitude (95 m here) and the GNSS/GPS altitude (149 m). The two differ in both reference and noise – the barometric value is referenced to a standard-atmosphere pressure setting and is smooth, the GPS value is absolute (relative to the WGS84 geoid) but vertically noisier – and on older loggers one of them may be missing (recorded as zero). Which altitude to use, and how, is left open and will be decided at the analysis stage (Chapter 4), because the vertical velocity derived from it drives the phase segmentation.

The sequence of `B` records is therefore a time-stamped 3-D trajectory. The sampling period is typically a few seconds but is not guaranteed to be uniform and varies with the recording device – a point the analysis stage must handle explicitly.

3.2 The catalog

Two derived tables make the raw files usable. The **catalog** (`catalog.csv`) has one row per flight: the competition metadata plus a `local_path` column linking the flight to its file on disk, so a flight can be traced to its track (and back, via the flight identifier embedded in the file name) without any external lookup. The **season index** (`seasons_index.csv`), versioned in the repository, summarizes each season with the number of flights declared, the number carrying a GPS track, and the number actually downloaded.

3.3 Coverage and statistics

The statistics below describe the FFVL CFD dataset currently in hand; further sources may be added later (Chapter 2). The dataset spans 27 seasons, from 1999–2000 to 2025–2026. It contains 203,343 flights in total; 186,225 of them (91.6 %) carry a GPS tracklog, and of those 186,052 (99.9 %) have been downloaded and verified. The per-season breakdown is given in Table 3.1. The volume of flights grows by more than three orders of magnitude over the period, reflecting the adoption of GPS loggers and online scoring: the early seasons contribute a handful of tracks each, while recent seasons contribute well over ten thousand.

Table 3.1: FFVL CFD flights per season: total declared, with a GPS track, and downloaded. Source: `data/seasons_index.csv`.

Season	Flights	With GPS	Downloaded	Coverage (%)
1999–2000	10	10	10	100.0
2000–2001	832	1	1	100.0
2001–2002	1,150	3	3	100.0
2002–2003	1,996	123	123	100.0
2003–2004	1,470	337	337	100.0
2004–2005	1,765	574	574	100.0
2005–2006	2,162	894	894	100.0
2006–2007	2,386	1,016	1,016	100.0
2007–2008	3,081	1,667	1,667	100.0
2008–2009	3,974	2,614	2,613	100.0
2009–2010	4,171	3,059	3,059	100.0
2010–2011	4,872	4,185	4,169	99.6
2011–2012	6,744	6,075	6,065	99.8
2012–2013	7,873	7,106	7,075	99.6
2013–2014	9,804	9,157	9,127	99.7
2014–2015	13,793	13,293	13,254	99.7
2015–2016	13,982	13,703	13,696	99.9
2016–2017	15,760	15,534	15,528	100.0
2017–2018	14,917	14,682	14,676	100.0
2018–2019	16,702	16,517	16,508	99.9
2019–2020	10,966	10,868	10,863	100.0
2020–2021	9,500	9,483	9,480	100.0
2021–2022	14,552	14,507	14,502	100.0
2022–2023	11,590	11,561	11,560	100.0
2023–2024	10,372	10,362	10,360	100.0
2024–2025	10,906	10,885	10,884	100.0
2025–2026	8,013	8,009	8,008	100.0
Total	203,343	186,225	186,052	99.9

3.4 Data quality and caveats

A few limitations are already known and will shape the analysis:

- **Flights without a track.** About 8 % of entries have no IGC file (older or manually declared flights); they carry metadata only.

- **A small residue of unrecoverable files.** A few flights advertise a track that the server does not actually serve as a valid IGC (broken or placeholder links); these are recorded as failures and excluded.
- **Heterogeneous logging.** Sampling rate, altitude source, and accuracy depend on the device; trajectories must be resampled and quality-filtered before any ensemble statistic is computed.
- **Incomplete dates.** Some historical entries have placeholder dates (e.g. `0000-00-00`); these are kept but flagged.

These caveats motivate the pre-processing described in Chapter 4.

Chapter 4

Next steps

With the dataset in place and verified, the work proceeds as follows.

4.1 Trajectory pre-processing

Each IGC file is parsed into a sequence of fixes $(t_i, \varphi_i, \lambda_i, h_i)$ – time, geodetic latitude and longitude, and altitude. The format records *two* altitudes per fix, a barometric (pressure) altitude and a GNSS (GPS) altitude; they differ in both reference and noise (the barometric value is smooth but tied to a pressure setting, the GPS value is absolute but vertically noisier). Which one to use – or whether to combine them – matters for the vertical velocity that drives the segmentation; this choice is left open for now and will be made at the analysis stage rather than fixed by default. The fixes are then resampled to a common cadence to absorb device heterogeneity, and quality-filtered (fixes with a bad validity flag, implausible speeds, or large time gaps are dropped). The output is a tidy, analysis-ready representation produced by a future `soaring.analysis` sub-package.

From geographic to local Cartesian coordinates. Latitudes and longitudes are angles on a curved surface and cannot be differenced directly to obtain distances; they are mapped to a local Cartesian frame in two steps, following the convention of Vilpellet et al. First, each fix is converted to Earth-Centred, Earth-Fixed (ECEF) coordinates on the WGS84 ellipsoid (semi-major axis $a = 6{,}378{,}137\,\mathrm{m}$, flattening $f = 1/298.257223563$, $e^2 = f(2-f)$),

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2\varphi}}, \qquad \begin{aligned} X &= (N + h)\cos\varphi\cos\lambda, \\ Y &= (N + h)\cos\varphi\sin\lambda, \\ Z &= \big(N(1 - e^2) + h\big)\sin\varphi, \end{aligned} \tag{4.1}$$

where $N(\varphi)$ is the prime-vertical radius of curvature. Second, taking the first fix $(\varphi_0, \lambda_0, h_0)$ of the flight as the origin, the ECEF displacement $\Delta\mathbf{X} = \mathbf{X} - \mathbf{X}_0$ is rotated into the local East–North–Up (ENU) frame tangent to the ellipsoid at that point,

$$\begin{pmatrix} E \\ N \\ U \end{pmatrix} = \begin{pmatrix} -\sin\lambda_0 & \cos\lambda_0 & 0 \\ -\sin\varphi_0\cos\lambda_0 & -\sin\varphi_0\sin\lambda_0 & \cos\varphi_0 \\ \cos\varphi_0\cos\lambda_0 & \cos\varphi_0\sin\lambda_0 & \sin\varphi_0 \end{pmatrix} \Delta\mathbf{X}. \tag{4.2}$$

The ENU frame *is* the local Cartesian frame: its horizontal plane (E, N) is the “north–east” plane and U is the local vertical, so there are not two distinct coordinate systems but one, reached through the intermediate (global) ECEF representation. Over the spatial extent of a single flight (tens of kilometres) the tangent-plane approximation is accurate to well below the GPS noise.

Notation. Throughout, $\mathbf{r}(t) = (x(t), y(t)) = (E(t), N(t))$ denotes the *horizontal* position in this local frame and $z(t) = U(t)$ the altitude. By construction each trajectory starts at the origin, $\mathbf{r}(0) = \mathbf{0}$, with the clock reset to $t = 0$ at take-off, so t is the elapsed flight time. The horizontal velocity is $\mathbf{v}(t) = \dot{\mathbf{r}}(t)$ and the horizontal displacement over a lag t is $\Delta\mathbf{r}(t) = \mathbf{r}(t_0 + t) - \mathbf{r}(t_0)$.

4.2 Observables on the whole trajectory

A first group of observables requires *no* segmentation: it can be measured directly on the full trajectories and already probes the transport at the global level. For each, we state what it is, how it is computed, and what it is for.

Transport and diffusion.

Mean-squared displacement (MSD). With every flight starting at the origin ($\mathbf{r}(0) = \mathbf{0}$), the *ensemble-averaged* MSD is the mean squared horizontal distance reached after an elapsed time t,

$$\mathrm{MSD}(t) = \langle\, |\mathbf{r}(t)|^2 \rangle, \tag{4.3}$$

where $\langle \cdot \rangle$ is an average *over flights* (an ensemble average) at fixed t. Its growth law $\mathrm{MSD}(t) \sim t^\alpha$ fixes the transport regime ($\alpha = 1$ Brownian, $\alpha > 1$ super-diffusive, $\alpha < 1$ sub-diffusive; equivalently the Hurst exponent $H = \alpha/2$). This is the primary test for anomalous transport.

Time-averaged MSD (TA-MSD). A second MSD estimator is built entirely *within* a single flight. For flight k of duration T_k, one slides a window of width t along the trajectory $\mathbf{r}^{(k)}$ and averages the squared displacement over all starting times t_0 *inside that flight*:

$$\overline{\delta^2}^{(k)}(t) \;\equiv\; \frac{1}{T_k - t} \int_0^{T_k - t} \left|\mathbf{r}^{(k)}(t_0 + t) - \mathbf{r}^{(k)}(t_0)\right|^2 \mathrm{d}t_0. \tag{4.4}$$

The average over t_0 here is a *time* average internal to flight k; it has nothing to do with the ensemble average $\langle \cdots \rangle$ over different flights used in the MSD. Because a single-flight curve $\overline{\delta^2}^{(k)}(t)$ is noisy, the operationally useful quantity is its mean over all flights of comparable duration T:

$$\mathrm{TAMSD}(t, T) \;\equiv\; \left\langle \overline{\delta^2}^{(k)}(t) \right\rangle_{T_k \approx T}. \tag{4.5}$$

Ergodic case. For an ergodic process, $\mathrm{TAMSD}(t, T) \to \mathrm{MSD}(t)$ as $T \to \infty$, independently of T, and the individual values $\overline{\delta^2}^{(k)}(t)$ concentrate around the common limit (the distribution collapses to a delta function).

Non-ergodic case: two independent signatures.

(i) *Aging.* $\mathrm{TAMSD}(t, T)$ differs from $\mathrm{MSD}(t)$ and retains a systematic dependence on T: flights of different duration yield different curves. This is a strong indicator of non-ergodicity, but it is sufficient, not necessary – some non-ergodic processes show only a weak T-dependence.

(ii) *Non-self-averaging.* The distribution of $\overline{\delta^2}^{(k)}(t)$ across trajectories remains broad even as $T \to \infty$: individual flights give systematically different TA-MSD values and the distribution does not collapse. This is the more fundamental signature: it persists even when aging is weak.

We therefore monitor both TAMSD(t, T) and the full distribution of $\overline{\delta^2}^{(k)}(t)$ across trajectories at fixed t and T. The canonical illustration – a CTRW with power-law waiting times – is worked out analytically in Appendix A, where both EA-MSD≠TA-MSD and the T-dependence are derived explicitly.

Propagator. Since each flight starts at the origin, $\mathbf{r}(0) = \mathbf{0}$, the position at time t and the displacement from the start coincide: $\mathbf{r}(t) = \mathbf{r}(t) - \mathbf{r}(0)$. We therefore define the propagator as the probability density of the position vector $\mathbf{x} \in \mathbb{R}^2$ at time t,

$$P(\mathbf{x}, t)\, \mathrm{d}^2 x = \Pr\big(\mathbf{r}(t) \in [\mathbf{x},\, \mathbf{x} + \mathrm{d}\mathbf{x}]\big). \tag{4.6}$$

For a *self-similar* process with exponent H the propagator obeys the scaling form

$$P(\mathbf{x}, t) = t^{-2H}\, G\left(\frac{\mathbf{x}}{t^H}\right), \tag{4.7}$$

where $G : \mathbb{R}^2 \to \mathbb{R}_{>0}$ is a time-independent scaling function (normalised, $\int G\, \mathrm{d}^2 u = 1$).

In practice one works with the **one-dimensional marginal** obtained by projecting onto a single Cartesian component x,

$$P(x, t) \equiv \int_{-\infty}^{+\infty} P((x, y), t)\, \mathrm{d}y. \tag{4.8}$$

Assuming the process is *isotropic*, $G(\mathbf{u}) = G(|\mathbf{u}|)$, the integral can be evaluated by the substitution $y = t^H v$:

$$P(x, t) = t^{-2H} \int_{-\infty}^{+\infty} G\left(\sqrt{\left(\tfrac{x}{t^H}\right)^2 + v^2}\right) t^H\, \mathrm{d}v = t^{-H}\, F\left(\frac{x}{t^H}\right), \tag{4.9}$$

where $F(u) \equiv \int_{-\infty}^{+\infty} G(\sqrt{u^2 + v^2})\, \mathrm{d}v$ inherits the normalisation $\int F\, \mathrm{d}u = 1$ from G. The marginal therefore satisfies the same scaling form as the 2D propagator, with the same exponent H but in one dimension. The shape of F distinguishes transport regimes: a Gaussian F characterises ordinary diffusion, while heavy power-law or stretched-exponential tails signal anomalous transport.

The marginal $P(x, t)$ is the most practical object to estimate from data and to collapse onto the scaling form (4.9) by rescaling the axis by t^H.

Isotropy is an assumption, not a given, and must be tested. Competition flights have preferred directions (race goal, orography, prevailing winds), so isotropy may fail. Four complementary tests are available directly from the data:

(i) *Variance ratio.* For an isotropic process $\mathrm{Var}(E(t)) = \mathrm{Var}(N(t)) = \mathrm{MSD}(t)/2$ for all t. The ratio $\mathrm{Var}(E(t))/\mathrm{Var}(N(t))$ should be identically 1.

(ii) *Marginal comparison.* The distributions $P(E(t) = x)$ and $P(N(t) = x)$ should coincide for all t; a Kolmogorov–Smirnov test provides a formal check.

(iii) *Cross-covariance.* For an isotropic process $\mathrm{Cov}(E(t), N(t)) = 0$; a non-zero value signals a preferred direction.

(iv) *PCA of the covariance matrix.* Diagonalise $\Sigma(t) = \langle \mathbf{r}(t)\mathbf{r}(t)^\top \rangle$. The eigenvalue ratio $\lambda_1(t)/\lambda_2(t)$ measures anisotropy ($= 1$ if isotropic); the eigenvectors reveal the dominant axis.

If isotropy fails, the derivation above is no longer valid: the two components may have different scaling functions or even different exponents. In that case one works with each component separately, or rotates to the principal axes of $\Sigma(t)$ before applying the scaling collapse.

Probability of return to the origin. The value of the propagator at its centre, $P_0(t) \equiv P(\Delta\mathbf{r} = \mathbf{0}, t)$, i.e. the height of the peak. For a self-similar process it decays as a power law set by the same exponent, $P_0(t) \sim t^{-dH}$ in d dimensions (t^{-2H} in the plane, t^{-H} for a single component). It therefore yields H from the *bulk* of the distribution rather than from its tails, which is valuable when heavy tails make the variance – and hence the MSD – ill-defined; a change of slope in $\log P_0$ versus $\log t$ exposes a crossover between regimes, and a peak exponent that disagrees with the MSD exponent is itself informative, pointing to multi-scaling. (This is the route Mantegna and Stanley used to read off the index of Lévy-like walks.)

Non-Gaussianity. How far the propagator departs from a Gaussian, measured by the non-Gaussian parameter, in the plane

$$\alpha_2(t) = \frac{\langle|\Delta\mathbf{r}(t)|^4\rangle}{2\,\langle|\Delta\mathbf{r}(t)|^2\rangle^2} - 1, \tag{4.10}$$

which vanishes for a Gaussian and is positive for heavier-than-Gaussian tails. It is evaluated on $P(\Delta\mathbf{r}, t)$ *at each lag* t: it is therefore a function of t, one number per lag, tracing a curve $\alpha_2(t)$ that pinpoints the time scales over which the displacement statistics are non-Gaussian (typically large at the intermediate lags where occasional long glides dominate, and decaying if a central-limit-type Gaussianisation sets in at long lags). It is a compact summary of the shape of P, not a substitute for it.

Velocity autocorrelation $C(\tau)$. $C(\tau) = \langle\mathbf{v}(t_0)\cdot\mathbf{v}(t_0+\tau)\rangle$ (often normalised by $C(0)$) is the dot product of the velocity with its later self: it is large and positive when, a delay τ later, the motion still points the same way, so the scale over which $C(\tau)$ decays *is* the memory time of the heading. It controls the MSD through the Green–Kubo relation $\mathrm{MSD}(t) = 2\int_0^t (t-\tau)\,C(\tau)\,\mathrm{d}\tau$: a slowly decaying C produces a faster-growing MSD. We do not presume the motion ends in ordinary diffusion; rather, the functional form of $C(\tau)$ is the diagnostic – an integrable C relaxes to normal diffusion, a non-integrable (e.g. power-law) tail sustains persistent super-diffusion with no crossover, and a sign change reveals the anti-persistence expected from circling.

Turning-angle distribution. The distribution of the angle θ between two successive horizontal displacement vectors along the trajectory – the change of heading from one step to the next. A distribution peaked at $\theta = 0$ means strong forward persistence (a correlated random walk); a flat distribution means memoryless reorientation. This is the microscopic origin of the persistence seen in $C(\tau)$ and the MSD.

First-passage and first-exit times. The first-passage time $T(L)$ is the first time the horizontal distance from take-off reaches a value L (the first-exit time, the first time the trajectory leaves a disc of radius L). For a self-similar transport process the characteristic time scales with the length as $\langle T(L)\rangle \sim L^{1/H}$, with dynamic exponent $z = 1/H$, so the slope of $\log\langle T\rangle$ against $\log L$ gives an estimate of H obtained in the space domain, independent of the MSD; the full distribution of $T(L)$ additionally exposes heavy tails coming from long trapping events.

Speed and vertical-velocity distributions. The magnitude of the horizontal velocity and the vertical velocity. Basic kinematic characterization; the vertical velocity also separates climbing from sinking and feeds the segmentation below.

Trajectory geometry?????

Tortuosity. How much a path winds, quantified by the ratio between the length actually travelled and the straight-line distance between its endpoints over the same interval (1 for a straight line, larger for a more convoluted path). It is a geometric *efficiency* descriptor and does not map one-to-one onto a transport exponent – very different dynamics can share a tortuosity – so we use it descriptively and to separate phases (close to 1 in glides, large in search and climb) rather than as an estimator of H.

Fractal dimension. The box-counting dimension d_f of the trail, i.e. the exponent with which the number of boxes of size ϵ needed to cover the path grows as $\epsilon \to 0$. For a self-similar walk it is tied to the transport exponent by

$$d_f = \min(1/H,\, 2), \tag{4.11}$$

so a ballistic line ($H = 1$) gives $d_f = 1$, a Brownian path ($H = 1/2$) gives $d_f = 2$, and the reported sub-ballistic soaring ($H \approx 0.88$) gives $d_f \approx 1.14$. The fractal dimension is thus the geometric counterpart of H and a third, independent route to it, after the MSD slope and the return/first-passage scalings; agreement among the three is a strong internal check, disagreement is evidence of multi-scaling.

Radius of gyration. The root-mean-square distance of the trajectory points from their centroid, i.e. the characteristic spatial extent of a flight. As a single number it sets the natural length scale of a flight (useful to normalise other observables); measured as it grows with elapsed time, $R_g(t) \sim t^H$, it offers yet another reading of the same exponent.

One number per flight, or a function of scale? The geometric quantities above can be read in two complementary ways. As a *single number per flight* – one tortuosity, one d_f, one R_g – whose distribution over the ensemble characterises the population and feeds the stratified comparisons. Or as a *function of scale or elapsed time* – $R_g(t)$ as the trajectory unfolds, tortuosity measured on sub-segments of growing duration t, the box count as a function of box size – which is precisely where the transport exponents live ($R_g(t) \sim t^H$; the box-count slope is d_f). The scalar is descriptive; the scale dependence is what links geometry back to the anomalous exponent, and both views will be used.

4.3 Segmentation into flight phases

Building on the previous work of Vilpellet et al. [4], each flight is segmented into the three phases of the soaring cycle:

Transition (glide). The inter-thermal cruise: a near-straight, energy-efficient glide toward the next source of lift (sinking, large directed horizontal displacement). This is the directed "step".

Search. The exploratory phase entered when no thermal is found at a comfortable altitude: gentle turns and probing manoeuvres to sample the air mass and locate the next thermal while limiting altitude loss. It mediates the passage from one step to the next trap.

Climb (thermalling). Circling within rising air to regain altitude: persistent turning with small net horizontal displacement. This is the localized "trap".

That work performs the segmentation with a hidden Markov model whose observations are a small set of *binary* trajectory-derived features — the sign of the vertical speed, the persistence of the curvature angle, and indicators of trajectory straightness.

A first task of this thesis is to scrutinize this segmentation rather than take it for granted: to assess whether those binary variables are adequate and robust on the larger, more heterogeneous dataset assembled here, and whether the feature set or the model should be refined, extended, or replaced. The segmentation method is therefore itself an object of study, and the specific variables are deliberately left open at this stage. This three-state description refines the two-state caricature underlying continuous-time random walks and Lévy walks.

4.4 Phase-resolved observables

The segmentation unlocks a second group of observables, defined per phase occurrence and then as ensemble distributions. These are the building blocks of an intermittent-transport description and, unlike those of Section 4.2, cannot be obtained without first segmenting the flight.

The waiting phase: climb and search. Between two consecutive glides the flier advances little horizontally: it searches for a thermal and climbs in it. For a transport model these two phases together form the *waiting time* between directed steps, so we treat them on an equal footing and build three distributions – the climb duration τ_c, the search duration τ_s, and their sum $\tau_w = \tau_s + \tau_c$, the total time spent essentially stationary between glides. The distinction is not cosmetic: Vilpellet et al. [4] report the search times to be broadly *power-law* distributed while the climb times are closer to *exponential*, so that the heavy tail of the combined waiting time $\psi(\tau_w)$ – the quantity that actually decides whether trapping drives sub-diffusion – would be inherited from the search phase. Whether this still holds on the larger dataset is itself one of the things to re-check once the segmentation is in place. We therefore report $\psi(\tau_w)$ together with the separate $\psi(\tau_c)$ and $\psi(\tau_s)$. Alongside the durations we record the climb kinematics (altitude gained Δz, mean climb rate, thermalling radius and period) and the search geometry (path length, tortuosity, net displacement) – the search being the phase in which prior work locates the signature of pilot skill.

The directed step: glide (transition). Each glide contributes a horizontal step of length ℓ and duration τ_g, with ground speed, glide ratio (horizontal distance per unit altitude lost) and sink rate. Because a glide is close to ballistic, $\ell \approx v_g \, \tau_g$, step length and duration are expected to be nearly proportional; we test this directly through the joint law of (ℓ, τ_g) – the ℓ–τ_g scatter, the conditional mean $\langle \ell \mid \tau_g \rangle$, and the spread of the ratio ℓ/τ_g (the glide speed). Both marginals matter: the tail of the step-length distribution $\lambda(\ell)$ controls super-diffusive contributions, so we compute the distributions of ℓ and τ_g separately as well as their joint law.

Anatomy of a flight. Number of thermals (climbs) per flight, inter-thermal distances, and the fraction of both time and along-track distance spent in each phase: the coarse composition that sets the weight of each phase in the global transport.

Angular diffusion between glides. The turning angle between the heading of one transition segment and the next – the segment-level counterpart of the turning angle of Section 4.2. How fast successive glide directions decorrelate is the key ingredient of the minimal model below.

Two couplings then decide which transport model applies, and are stated here as explicit objects of study. The first is the *step–waiting coupling*: whether a long glide tends to be followed by a long wait, i.e. whether ℓ and τ_w are correlated. A genuine coupling is the defining feature of a Lévy walk and is what separates it from a decoupled CTRW, in which jumps and waiting times are drawn independently.

The second is the *memory in the phase sequence*. A flight is a discrete sequence over the three states $\{\text{transition}, \text{search}, \text{climb}\}$, and memory can hide in two places. (i) The *order* of the phases, summarised by the transition matrix $P_{ij} = \Pr(\text{next} = j \mid \text{current} = i)$: a perfectly cyclic flight ($\mathrm{T} \to \mathrm{S} \to \mathrm{C} \to \mathrm{T}$) gives a near-deterministic matrix, whereas aborted climbs, repeated searches or skipped phases appear as off-cycle probabilities and reveal how stereotyped the soaring cycle actually is. (ii) The *durations*, through correlations between consecutive phase durations – for instance whether a long climb tends to precede a long glide. Both probe the *renewal* assumption underlying the textbook CTRW and Lévy-walk results, in which each step and wait is independent of the history: if the transition matrix is structured or the durations are correlated, the process is non-renewal and those formulas must be amended. Measuring these is therefore what tells us whether a memoryless model is admissible at all.

4.5 Reproducing the reference results

A first objective of the analysis, undertaken once the segmentation is in place, is to check whether the enlarged—and eventually multi-source—dataset reproduces the quantitative and graphical results of Vilpellet et al. [4]. Two figures are the main targets.

The first is the per-phase conditional MSD (their Fig. 3): the transition segments should stay ballistic ($\alpha = 2$, i.e. $H = 1$) across time lags; the search phase should be strongly sub-diffusive ($H \approx 0.3$ for paragliders and hang gliders, ≈ 0.2 for sailplanes); and the climb phase should be quasi-ballistic at long lags but with an effective velocity about two orders of magnitude smaller, including the transient anti-correlation near the $\sim 30\,\mathrm{s}$ thermalling period.

The second is the set of per-phase descriptive statistics (their Fig. 4): the glide ratios (≈ 8.5, 10, and 25 for paragliders, hang gliders, and sailplanes), the horizontal-velocity statistics, the heavy-tailed distribution of transition times, the search-time fraction and the effective search radius, the vertical climb velocities, the thermalling radii, and the thin, near-exponential distribution of climb times—together with the global time budget ($\approx$60% transition, $\approx$35% climb, the remainder searching) and the global sub-ballistic scaling ($H \approx 0.88$).

Recovering these results would validate the acquisition pipeline and the segmentation and would test the universality of the reported behaviour on a substantially larger sample; systematic departures would be equally informative, pointing to dataset- or method-dependent effects to investigate. This comparison is therefore the bridge between the present data-acquisition stage and the original analysis that follows.

4.6 Transport analysis

Using these observables, test for anomalous transport: estimate the MSD scaling exponent α, fit the tails of $\psi(\tau)$ and $\lambda(\ell)$ with principled methods (e.g. maximum-likelihood power-law fits with goodness-of-fit tests), and perform model selection between the continuous-time random walk and the Lévy-walk frameworks [1, 2] – using the step/waiting-time coupling to tell them apart. Because logging devices, sites, gliders and weather differ, results will be stratified (by season, site, glider class, and data source) and trajectories normalized, and finite-size, censoring and sampling biases will be controlled throughout.

4.7 Modelling and simulation

The aim is a *minimal* stochastic model that reproduces the asymptotic anomalous scaling reported for soaring (a Hurst exponent $H \approx 0.88$) with as few ingredients as possible. A preliminary working hypothesis—to be made precise, and possibly revised, once the empirical phase statistics are available—is that the transport is carried mainly by the transition (glide) phases. The model would therefore resolve the transition segments explicitly while merging search and climb into a single effective non-transition state, so that the soaring cycle reduces to a sequence of directed glides interrupted by reorientations. Anomalous transport would then emerge from the directed transition segments combined with an *angular diffusion* of the heading from one transition to the next: successive glide directions decorrelate gradually rather than abruptly, producing a correlated random walk whose persistence controls the long-time exponent. Simulated ensembles would be compared with the data to test whether these ingredients suffice. These are baseline ideas, to be refined—or changed—as the analysis progresses.

4.8 Tooling

Mirror the acquisition package with an analysis package, and let the figures and tables of this document be generated automatically from the analysis outputs – as the dataset statistics already are – so that this document stays in step with the work and remains a reliable basis for the thesis and for presentations.

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Appendix A

CTRW with power-law waiting times: EA-MSD vs TA-MSD

This appendix derives the ensemble-averaged MSD (EA-MSD) and the time-averaged MSD (TA-MSD) for the continuous-time random walk (CTRW) with heavy-tailed waiting times, and shows explicitly that the two differ and that the TA-MSD depends on the observation time T.

Model

A walker takes steps of length l_i (iid, symmetric, $\langle l^2 \rangle = \sigma^2 < \infty$) separated by waiting times τ_i (iid) drawn from

$$\psi(\tau) \sim \frac{\alpha\, \tau_0^\alpha}{\tau^{1+\alpha}}, \qquad \tau \to \infty, \quad 0 < \alpha < 1. \tag{A.1}$$

The mean waiting time diverges ($\langle \tau \rangle = \infty$), so the walker is anomalously slow. Let $n(t)$ be the number of steps completed by time t, $T_n = \sum_{i=1}^{n} \tau_i$ the time of the n-th step, and

$$x(t) = \sum_{i=1}^{n(t)} l_i \tag{A.2}$$

the position at time t.

Ensemble-averaged MSD

Since steps and waiting times are independent,

$$\langle x^2(t) \rangle = \sigma^2\, \langle n(t) \rangle. \tag{A.3}$$

For large t the mean number of renewals in $[0, t]$ follows from the renewal equation and the Tauberian theorem applied to the Laplace transform of ψ. Because $\hat{\psi}(s) \equiv \int_0^\infty e^{-s\tau} \psi(\tau)\, \mathrm{d}\tau \simeq 1 - (\tau_0 s)^\alpha + \cdots$ for $s \to 0$ (a consequence of ψ having a power-law tail with exponent $1 + \alpha$), the renewal theorem gives

$$\langle n(t) \rangle \sim \frac{t^\alpha}{\tau_0^\alpha\, \Gamma(1+\alpha)}, \qquad t \to \infty. \tag{A.4}$$

Hence the **EA-MSD** grows sub-linearly,

$$\mathrm{MSD}(t) = \langle x^2(t) \rangle \sim \frac{\sigma^2}{\tau_0^\alpha\, \Gamma(1+\alpha)}\, t^\alpha \equiv K_\alpha\, t^\alpha, \qquad 0 < \alpha < 1. \tag{A.5}$$

The transport is sub-diffusive with Hurst exponent $H = \alpha/2 < 1/2$.

Time-averaged MSD and its mean

For a trajectory of duration T, the time-averaged MSD at lag $t \ll T$ is

$$\overline{\delta^2}(t;T) = \frac{1}{T-t}\int_0^{T-t} \left[x(t_0+t) - x(t_0)\right]^2 \mathrm{d}t_0. \tag{A.6}$$

Since $\langle [x(t_0+t) - x(t_0)]^2\rangle = \sigma^2 \langle n(t_0+t) - n(t_0)\rangle$, the mean of the integrand equals σ^2 times the mean number of steps in any window of width t. The key observation is that the integral $\int_0^{T-t}[n(t_0+t) - n(t_0)]\,\mathrm{d}t_0$ counts, for each step k that lands at time $T_k \in [0,T]$, the length of the t-wide sliding window that contains T_k. For T_k deep in the interior ($t \le T_k \le T-t$) this length is exactly t; the contributions from the two boundary layers of width t are negligible for $t \ll T$. Therefore

$$\int_0^{T-t} [n(t_0+t) - n(t_0)]\,\mathrm{d}t_0 \approx t\,n(T), \qquad t \ll T, \tag{A.7}$$

and

$$\overline{\delta^2}(t;T) \approx \frac{\sigma^2\, t\, n(T)}{T}. \tag{A.8}$$

Taking the ensemble average and using $\langle n(T)\rangle \sim K_\alpha T^\alpha/\sigma^2$,

$$\mathrm{TAMSD}(t,T) \equiv \langle \overline{\delta^2}(t;T)\rangle \approx \sigma^2\,\frac{\langle n(T)\rangle}{T}\,t \sim K_\alpha\, T^{\alpha-1}\, t. \tag{A.9}$$

Comparing (A.5) and (A.9):

- The EA-MSD scales as t^α (sub-diffusive), whereas the TAMSD scales as t^1 (linear in the lag) – a qualitatively different dependence on t.
- The TAMSD carries an explicit $T^{\alpha-1}$ prefactor: since $\alpha < 1$ this is a decreasing function of T, so longer flights yield smaller TAMSD at the same lag. This T-dependence is the signature of *aging*: the process has not forgotten its past, and the answer depends on how long we have observed it.
- The ratio $\mathrm{TAMSD}(t,T)/\mathrm{MSD}(t) \sim (t/T)^{1-\alpha} \to 0$ as $T \to \infty$ at fixed t – the two observables diverge.

Distribution of the TA-MSD: non-self-averaging

Equation (A.8) shows that $\overline{\delta^2}(t;T) \approx \sigma^2 t\, n(T)/T$, so the randomness in the TA-MSD comes entirely from $n(T)$, the total number of steps in $[0,T]$. For a power-law renewal process the rescaled count

$$\xi \equiv \frac{n(T)}{\langle n(T)\rangle} \tag{A.10}$$

converges in distribution as $T \to \infty$ to a non-degenerate limit law: the *Mittag-Leffler distribution* M_α, whose probability density function is defined through its moments $\langle \xi^n\rangle = n!\,\Gamma(1+\alpha)^n/\Gamma(1+n\alpha)$. This is a broad, non-Gaussian distribution on $(0,\infty)$ that does *not* concentrate around $\xi = 1$ for $\alpha < 1$: different trajectories give systematically different values of $\overline{\delta^2}(t;T)$ even in the limit $T \to \infty$. This is *non-self-averaging*: the TA-MSD of a single trajectory is not a reliable estimate of any ensemble quantity, regardless of how long the trajectory is.

Quantitatively, the ergodicity-breaking parameter

$$\mathrm{EB}(t,T) \equiv \frac{\langle [\overline{\delta^2}(t;T)]^2\rangle - \langle \overline{\delta^2}(t;T)\rangle^2}{\langle \overline{\delta^2}(t;T)\rangle^2} = \mathrm{Var}(\xi) \sim c_\alpha\, T^{-\alpha} \to 0 \tag{A.11}$$

does tend to zero (the variance of ξ decreases with T), but the *distribution* of ξ converges to a non-trivial limit rather than to a delta function. This is why monitoring the full distribution of $\overline{\delta^2}^{(k)}(t)$ across trajectories – not just its mean – is the proper diagnostic for non-ergodicity: the collapse (or non-collapse) of that distribution as T grows tells us whether we are dealing with a self-averaging ergodic process or a non-self-averaging non-ergodic one.

For comparison, an ordinary random walk ($\alpha = 1$, finite mean waiting time) has $n(T)/\langle n(T)\rangle \to 1$ in probability as $T \to \infty$, so the distribution of the TA-MSD collapses to a delta function, $\overline{\delta^2}(t;T) \to \mathrm{MSD}(t)$ almost surely, and the process is ergodic.